Induced Subgraphs in Ramsey Graphs

University of Oxford

Mathematical Institute

Examination Number: XXXXXXXX
Degree Course: MSc in Mathematical Sciences
Submission: Trinity Term 2026

Submitted in partial fulfilment of the requirements for the degree of Master in Mathematical Sciences (OMMS).

Word count: [**INSERT TEXCOUNT WORD COUNT HERE**]

Abstract

Replace this paragraph with a concise summary (100–200 words) of the dissertation: the problem studied, the methods used, and the principal results obtained.

Contents

1 Introduction **3**

2 Subgraphs with Many Degrees **4**

3 Subgraphs of Quadratically Many Sizes **5**

4 The Erdős–McKay Conjecture **6**

4.1 The Full Erdős–McKay Conjecture and Its Resolution 6
4.2 Quadratic Polynomials of Gaussians 6
4.3 The Carbery–Wright Theorem 6
4.4 Fourier Analysis and Characteristic Functions 6
4.5 Additive Structure Dichotomy 6
4.6 The Two-Scale Decomposition 6
4.7 The Buckets Argument 6

5 Conclusing Remarks **7**

A Supplementary Proofs **8**

1 Introduction

2 Subgraphs with Many Degrees

Definition 2.1 ((c, δ)-diversity)**.**

Theorem 2.2 (Bukh–Sudakov 2007 [])**.** *Every C-Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(n)$ distinct degrees.*

Proof. □

3 Subgraphs of Quadratically Many Sizes

Definition 3.1 ((δ, ε)-rich graph)**.**

Theorem 3.2 (Kwan–Sudakov 2020 [])**.** *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Proof. □

4 The Erdős–McKay Conjecture

4.1 The Full Erdős–McKay Conjecture and Its Resolution

4.2 Quadratic Polynomials of Gaussians

4.3 The Carbery–Wright Theorem

Theorem 4.1 (Carbery–Wright []).

4.4 Fourier Analysis and Characteristic Functions

4.5 Additive Structure Dichotomy

4.6 The Two-Scale Decomposition

4.7 The Buckets Argument

Theorem 4.2 (Kwan–Sah–Sauermann–Sawhney 2023 []).

5 Conclusing Remarks

References

A Supplementary Proofs